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January 17th, 2026

To the Selection Committee,

I am delighted to recommend Glenn Nurwono for the Distinguished Teaching Assistant Award. Since joining the Department of Chemistry in Fall 2021, Glenn has been an outstanding and impactful TA, contributing meaningfully to courses across Chemistry and Chemical Engineering. What distinguishes him is his sustained commitment to growth as an educator, his ability to engage students at all levels, and his initiative in taking on increasing instructional responsibility, most recently through the development of his own Clusters Seminar. He fosters supportive learning environments while serving as a proactive and reliable partner to instructors, making him an invaluable member of every teaching team.

I had the pleasure of working closely with Glenn during the inaugural offering of my Natural Product Biosynthesis course (CHEM 138/238). With no prior course materials, Glenn played a central role in shaping the class. Beyond his initial assessment duties, he designed weekly problem sets that offered formative practice and clear insight into student understanding. For exams, he developed thoughtful questions emphasizing critical thinking and engagement with primary literature, extending learning beyond lecture content and into real-world research. Glenn also excelled in direct student interactions, consistently answering questions, facilitating discussions, and guiding presentations. His contributions ensured the course ran smoothly and enriched the materials and structure adopted in subsequent iterations.

Glenn has demonstrated the same dedication and effectiveness across a wide range of instructional contexts. In the Molecular Biotechnology Laboratory (CH ENGR 104D), he engaged students with limited biology experience, refined lab modules, and integrated modern workflows to improve experiment success rates and reduce extraneous work. His improvements were significant enough that he was invited to teach the course again the following quarter, working with a different professor to help institutionalize these changes. Across introductory and upper-division courses, lectures and labs, Glenn consistently receives outstanding student feedback for making material approachable. These experiences highlight both his reliability as an instructor and his versatility across departments and diverse student populations.

What truly distinguishes Glenn is his thoughtful approach to teaching as a craft and his sustained initiative to grow beyond the classroom. He actively pursues pedagogical training in writing, inclusive teaching, and research mentorship, and applies these insights to improve his teaching practice. This commitment culminates in his independent design of a Clusters Seminar this spring,

which emphasizes science communication for an interdisciplinary student audience. His initiative to seek out and integrate these opportunities alongside his research and teaching reflects a deep dedication to undergraduate education and a clear ability to implement student-centered pedagogical approaches.

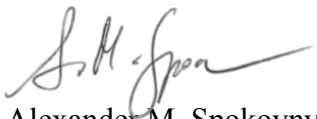
In summary, Glenn is a thoughtful and highly effective educator. His commitment to accessibility, adaptability, and intentional course design is evident across his diverse teaching experiences, as he continues to grow as an educator both inside and outside the classroom. His ability to respond to student needs, foster welcoming learning environments, and contribute meaningfully to course development makes him an exceptional candidate for the Distinguished Teaching Assistant Award. I recommend Glenn Nurwono for this honor without reservation.

Sincerely,



Yi Tang

I support this nomination:



Alexander M. Spokoyny
Professor and Department Chair
UCLA Chemistry & Biochemistry
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Glenn Nurwono – Nominee Statement

Since Fall 2021, I have taught eight courses across the curriculum, serving students from diverse academic backgrounds and career aspirations. My teaching trajectory reflects a sustained commitment to pedagogical excellence through three interconnected dimensions: developing instructional practices across varied teaching contexts, pursuing professional development, and contributing to the broader teaching community. Across all three, I have created inclusive, engaging environments where students at all stages can be inspired and challenged.

Teaching Across Diverse Contexts and Student Populations

From first-year chemistry laboratories to graduate-level electives, my teaching spans disciplines, course levels, and instructional formats. These experiences have shaped a student-centered approach that adapts to academic stage, prior preparation, and confidence. In CHEM 14BL, a first-year lab taught during one of the first fully in-person quarters after COVID-19, many students had limited laboratory experience and heightened anxiety. To support them, I developed visual guides, procedural demonstrations, and stepwise scaffolds that clarified expectations (Artifact 3). Established community agreements also normalized questions and not understanding at first. Students described the lab as a supportive space, as these interventions built foundational confidence and habits that students carried into upper-division courses.

In upper-division lecture courses, where graduate study and careers are more immediate, I emphasize disciplinary reasoning and real-world application. In CHEM 153A and CHEM 138/238, I designed practice worksheets, exams, and literature-based assignments that emphasized synthesis, critical thinking, and engagement with primary research (Artifact 4). These low-stakes opportunities allowed students to apply concepts to novel problems while scaffolding complexity to maintain engagement and foster curiosity. These experiences helped students identify interests that guided future coursework and careers.

In upper-division laboratory courses, I focus on integrating conceptual understanding with practical, transferable skills. In the senior-level Molecular Biotechnology Laboratory (CH ENGR 104D), I modernized lab protocols and introduced computational workflows that aligned better with industry-relevant skills (Artifact 3). In CHEM 153L, an upper-division biochemistry lab, I emphasized formative feedback and discipline-specific scientific writing, helping students interpret data, communicate results, and revise their work, essential for research and industry. Together, these experiences, along with consistently positive student evaluations (Artifact 5), illustrate a cohesive approach that prepares students across academic stages, departments, and instructional contexts.

Professional Development and Evidence-Driven Teaching

To refine my practice, I have pursued professional development spanning macro-level course design to micro-level instructional strategies. The Preparing Future Faculty Program introduced teaching as a scholarly, evidence-based practice, including course design and pedagogical literature. CIRTL Certification emphasized inclusive pedagogy, active learning, and teaching-as-research (TAR), culminating in my TAR project investigating how scaffolded writing and explicit connections between science and students' everyday lives affect engagement and confidence among non-STEM students (Artifact 2).

The Graduate Writing Certificate Program further shaped my approach, deepening my understanding of writing as a tool for learning. Through scaffolded assignments, transparent criteria, and reflective feedback, I implemented these practices in my courses to help students clarify thinking, synthesize knowledge, and monitor progress. This also informs Clusters

Seminar I designed, where interdisciplinary content and structured writing instruction will help students from diverse majors engage meaningfully with science in personal, societal, and ethical contexts (Artifact 1). These experiences reflect a deliberate progression from course design to daily instructional choices, grounded in evidence and focused on improving learning.

Contributions to the Broader Teaching Community

Learning occurs within a community, and I actively contribute through mentorship and service. I guide first-year graduate TAs in microteaching during orientation, provide feedback to undergraduate Learning Assistants in the Science Education Minor, and foster reflective, student-centered instruction. Beyond the classroom, I serve as a peer mentor in the IGNITE program, supporting first-year graduate students in coursework, research, and teaching. Through IGNITE and mentoring undergraduates in my lab, I also apply evidence-based strategies learned from the Entering Mentoring Training Program. As a member of the Chemistry Department's Diversity Leadership Committee, I help organize events to foster increased sense of belonging for graduate students and postdocs. Outside the department, I have served on the Exploring Your Universe organizing committee every year of my graduate career, providing hands-on science experiences to K-12 students, from virtual programming during the pandemic to in-person events that most recently attracted over 12,000 visitors.

These efforts have been recognized with multiple awards, including the Michael E. Jung Excellence in Teaching Award, the Ralph & Charlene Bauer Award, the Herbert Newby McCoy Mentorship Award, and the Excellence in Science Communication Award. Across courses, departments, and outreach, my teaching demonstrates sustained commitment to student-centered learning, evidence-driven practice, and community engagement.

Glenn Nurwono – Statement of Teaching Philosophy

I approach teaching as the development of an interconnected system of thought that enables students to transfer knowledge across contexts. Undergraduates are not here just to pass classes, but to prepare for future careers. My role is therefore not to focus on undergraduate education as an endpoint, but to help them see the value of the material in the context of their future aspirations and to equip them to tackle challenges in their professional environments.

Creating Inclusive and Supportive Learning Environments

Effective learning begins where students are. Through teaching a wide range of students, I have found that while learning activities and strategies may differ, students engage most effectively when they perceive instructors as supportive and feel comfortable asking questions. I therefore prioritize creating approachable learning environments where students feel safe to ask questions and make mistakes, encouraging deeper engagement with course material rather than disinterest (Artifact 5). Clear communication, transparent expectations, and patient guidance reduce unnecessary anxiety in unfamiliar settings, whether learning new techniques in laboratory courses or grappling with new theories in lecture.

Formative feedback is central to this approach. I believe assignments should not be merely mechanisms for grading, but also opportunities to provide low-barrier, iterative opportunities for students to practice reasoning, reflect on their understanding, and receive guidance. This is particularly important in STEM courses, where opportunities for "first drafts" are rare and assessments are often summative. I provide additional practice problems and formative tasks that allow me to gain insight into students' thought processes and assess understanding independent of punitive grading (Artifact 4AB). This feedback also goes two ways, as mid-quarter evaluations give students a voice in shaping their learning and my teaching practices, supporting reflective instruction. By cultivating a sense of belonging and offering low-stakes pathways for assessment, I encourage students to engage authentically and confidently with their learning.

Designing for Transferable Learning and Disciplinary Thinking

Central to this goal is helping students see their learning as extending beyond a single course, making connections that are relevant for future classes and career aspirations. This involves guiding students in how to learn: connecting ideas across contexts to adapt their thinking. I design courses so that learning goals, activities, and assessments function as an interconnected system that fosters transferable skills. Using backward design, I identify the reasoning processes students should develop and scaffold instruction toward those outcomes. Summative assessments are structured not to measure recall, but to encourage disciplinary thinking, pattern recognition, and synthesis of foundational knowledge. Exam problems are grounded in real-world contexts, preparing students to apply their reasoning flexibly in unfamiliar situations while connecting back to core principles (Artifact 4CD).

Because transferability also encompasses practical skills, I also help students develop techniques and concepts relevant to their future careers. Drawing on my experience in research, I

have modernized lab courses to include relevant, contemporary skills that enhance learning and provide tangible competencies for future work (Artifact 3A-D). Teaching multiple courses across levels and departments has strengthened my ability to identify core concepts and design learning experiences that equip students to apply their knowledge broadly.

Writing and Communication as Tools for Learning

Writing plays a central role in how I support student learning and knowledge transfer. Because learning, like writing, is a process, I emphasize development and revision over product or grade alone. Intentionally structured, contextually relevant writing assignments not only assess understanding, but also promote metacognition and clear communication that support students in future applications.

Drawing on training in writing pedagogy, I scaffold writing assignments to make writing more approachable while integrating them alongside new disciplinary content, positioning communication as part of the learning process rather than an add-on. This approach informs the design of my Clusters course (Artifact 1), where assignments guide students in interpreting science and adapting communication for different audiences in the context of alternative proteins and their ethical, societal, and economic implications. Techniques such as reverse outlining, guided feedback, and rubric transparency make expectations explicit and emphasize revision. By framing writing as a process of inquiry rather than a final product and using interdisciplinary content that engages students from diverse majors, the course supports students in developing confidence and adaptability as learners for transfer to future contexts.

Although many science and engineering courses offer limited opportunities to use writing as a learning tool, I intentionally create these opportunities in the courses I support. In upper-division chemistry courses, I facilitate journal-club–style presentations that require students to situate course concepts within current research and communicate complex information to an intentional audience. Reflective writing on ethical dimensions in engineering courses such as Molecular Biotechnology further encourages students to connect technical knowledge with broader societal implications. Through these activities, students develop transferable strategies, including organizing complex information and revising work iteratively. By integrating writing with research engagement and real-world contexts, I prepare students for advanced coursework and professional practice.

Commitment to Growth as an Educator

As an instructor, I believe that while I guide students in learning, I am also continually learning how to teach. Teaching is an evidence-driven, reflective practice that requires ongoing refinement. Professional development, student feedback, and systematic inquiry into teaching effectiveness inform how I adapt instructional strategies (Artifact 2,3). This iterative process ensures that my teaching not only meets the unique learning demands of each course but also evolves with new pedagogical approaches to better equip students.

By designing inclusive learning environments, emphasizing formative feedback, fostering transferable reasoning, and using writing as a tool for growth, my philosophy centers on helping

students learn how to learn. Teaching students at different stages of their academic careers, both within STEM and across other disciplines, underscores the importance of helping all learners see how they can engage meaningfully with science and apply course content in real-world and future contexts. My goal is for students to leave my courses with the conceptual tools, learning strategies, and confidence needed to adapt, communicate, and thrive well beyond graduation, wherever their paths may lead.

Teaching Artifact 2: Teaching-As-Research Proposal [Excerpt]

Note: This teaching artifact is an excerpt from my Teaching-as-Research (TAR) proposal developed through the CIRTL certification pathway. Treating teaching as an evidence-driven, iterative practice, I identify a focused research question on student learning, design an instructional intervention, and outline a plan to assess its effectiveness in my Clusters course.

Bridging the Gap: Non-STEM Students' Confidence and Engagement with Science

Introduction

Many students experience a disconnect between science and the humanities, often seeing science as objective facts and humanities as subjective interpretation. This divide can make emerging scientific topics feel inaccessible or irrelevant to those outside of STEM, which may contribute to disengagement from public discourse. Non-STEM majors also face additional inclusivity barriers in STEM education, as introductory courses without intentional connections to how science intersects with broader social, cultural, or personal contexts reduces student investment in the course[1]. Yet scientific knowledge and reasoning influence non-STEM students' daily lives just as much as STEM students', particularly when engaging with socio-scientific issues that require both scientific literacy and confidence in applying scientific thinking to real-world decisions.

This disconnect is particularly visible in the domain of emerging food technologies, such as alternative proteins, where public opinions on cultured meat and plant-based alternatives are influenced not only by scientific evidence but also by cultural norms, industry messaging, and ethical considerations. In the context of food, science and technology, cultural values, industry incentives, and ethical considerations are deeply intertwined. These intersections shape both students' personal food choices and broader debates around policy and regulation, providing an opportunity to examine how students engage with scientific content across genres and audiences.

This TAR project investigates how first-year writing students develop scientific literacy, curiosity, and confidence as non-STEM majors. I am especially interested in students' perceptions of science, its relevance to their lives, and their willingness to engage with scientific reasoning when the course intentionally integrates scientific, ethical, and cultural considerations. Existing literature indicates that engagement with science is shaped not only by content knowledge but also by perceptions of relevance, confidence, and curiosity, as well as by how information is framed, the genres in which it appears, and implicit assumptions about expertise and credibility[2,3]. Intentional course design that emphasizes socio-scientific issues, reflection, and interdisciplinary framing can enhance non-STEM students' interest in science, their confidence in evaluating evidence, and their curiosity to explore scientific questions in daily life[4,5].

Taken together, these studies suggest that integrating genre awareness, scaffolded writing assignments, guided reflection, and intentional connections between science and societal issues can help non-STEM students develop critical engagement with science, moving beyond rote technical knowledge. For my TAR project, I will investigate how these activities influence students' perceptions of science, their confidence in engaging with scientific questions, and their ability to apply scientific knowledge in every day and policy contexts. My research question will be addressed through two complementary lenses:

1. **Student engagement and perceptions:** How do students' barriers, anxieties, or confidence in interacting with scientific texts change following scaffolded instruction in genre, audience, and critical reading?
2. **Science relevance and communication:** How do students' beliefs about the importance and applicability of science in daily life and decision-making, as well as their perceived ability to interpret, communicate, and apply scientific information to non-expert audiences, shift over the course?

Course Context and Implementation

The Evolution of the Cosmos and Life Cluster is a year-long interdisciplinary course designed primarily for non-STEM first-year students. The program fulfills multiple General Education and writing requirements and culminates in a small seminar course on a special topic of the instructor's choice, which I will teach. This seminar will be a Writing II course that meets once per week and incorporates assigned readings, scaffolded writing assignments, and a final writing project. With an enrollment cap of 25 students, the course allows for active learning through writing and reading workshops, explicit instruction in scientific genres, and guided discussions that connect scientific topics to broader cultural, ethical, and everyday contexts. My intervention will emphasize helping students engage with scientific information as it appears across academic and public settings, while reflecting on its relevance beyond STEM disciplines. Assessment will include pre- and post-course surveys, along with ongoing open-ended reflective prompts, to examine how students' perceptions of science, confidence with scientific texts, and sense of relevance evolve over the quarter.

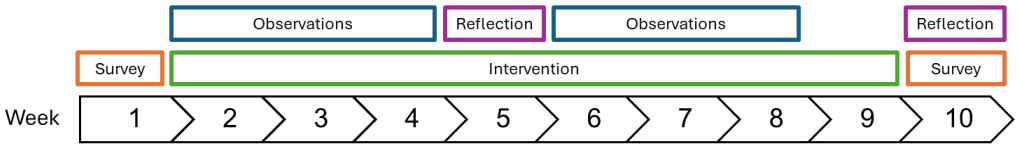


Figure 1: Project timeline

Teaching Artifact 2: Teaching-As-Research Proposal [Excerpt]

(Selected) Pre and Post-Survey Questions

Prompt: Below are a series of statements about reading scientific texts, connecting science to everyday life, and communicating scientific ideas. Please select the response that best reflects your current attitudes and confidence.

Four-point Likert scale: (1) Strongly Disagree, (2) Disagree, (3) Agree, (4) Strongly Disagree

I. Scientific Reading Confidence and Genre Understanding

Q1	I understand the typical structure of a scientific research article (e.g., Introduction, Methods, Results).
Q2	I can identify the main research question or knowledge gap a scientific paper is addressing.
Q3	I can recognize what evidence or data would strengthen a scientific argument.
Q4	I can generate meaningful questions about what future research could build on a study's findings.

II. Science Relevance and Real-World Connections

Q5	I see connections between scientific topics and everyday life decisions.
Q6	I understand how science is connected to social, ethical, or policy issues.
Q7	Scientific information influences how I think about current events or public debates.
Q8	People's values and backgrounds can shape how they interpret scientific information.

III. Writing and Communication Skills

Q9	I believe communicating science to non-experts is an important skill.
Q10	I view the revision process as important in my writing.
Q11	I can locate scientific sources to support my communication.
Q12	I can explain scientific ideas in ways that are clear to a general audience.

IV. Views on science

Q13	I see myself as someone who can understand scientific information.
Q14	I feel confident asking questions about scientific topics.
Q15	I feel confident participating in conversations about science-related issues.
Q16	I feel that science is relevant to my life and decisions.

(Selected) Open-Ended First Reflection Questions

- What has shaped your confidence (or lack of confidence) in engaging with science?
- Describe a time when science felt either accessible or inaccessible to you.
- When reading science, what kinds of questions do you find yourself asking?
- How do you decide whether a science-related news story is trustworthy?
- What two words come to mind when you think of scientific research? [Gauge during the course]

(Selected) Open-Ended Final Reflection Questions

- What is one way your understanding of science has changed during this course?
- What is one way your understanding of science communication has changed during this course?
- How has your confidence in reading scientific information changed over time?
- What skills from this course do you think you will use in the future?
- What two words come to mind when you think of scientific research? [Gauge after the course]

References

1. Smyth, D. S. *et al.* From BioSkills to LifeSkills—a framework for developing life skills for non-science majors in introductory life science courses. *Journal of Microbiology & Biology Education* **0**, e00002-26 (2026).
2. Lucas, K. L. Examining Science Identity Work and Scientific Literacy in Non-STEM Majors. (UC Santa Barbara, 2021).
3. Majetic, C. & Pellegrino, C. When Science and Information Literacy Meet: An Approach to Exploring the Sources of Science News with Non-Science Majors. *College Teaching* **62**, 107–112 (2014).
4. Takahashi, B. & TandocJr, E. C. Media sources, credibility, and perceptions of science: Learning about how people learn about science. *Public Underst Sci* **25**, 674–690 (2016).
5. Gormally, C. & Heil, A. A Vision for University Biology Education for Non-science Majors. *LSE* **21**, es5 (2022).

Teaching Artifact 3: Improving Learning and Accessibility in the Laboratory

CH ENGR 104D Syllabus and Course Improvements

Context: This upper-division elective lab course serves senior Chemical Engineering majors with limited biology backgrounds. I identified barriers to student success and implemented evidence-based improvements across lab design, course structure, and instructional materials. This artifact demonstrates initiative for course refinement based on student needs and intentional development of transferable skills applicable to both industry and academic research settings.

A)

6. **Important: in the evening before (consult with TA) Session 1, the group should fill two sterile test tubes with 5 mL LB medium (with Kanamycin added to a final concentration of 35 µg/mL), and inoculate the test tubes with a single colony of "pET28a_*lacZ*". Incubate the cultures until the next morning in the shaker at 37°C, 300 rpm. *Note: Kanamycin is a select carcinogen.***

7. **Important: 3-4 hours before Session 1 (consult with TA), the group should fill one sterile 500-mL shake flask with 100 mL LB medium (with Kanamycin at 35 µg/mL), and transfer 10 mL of the overnight culture to the shake flask with kanamycin. Incubate the cultures for 2 hr in the shaker at 37°C, 300 rpm, which will serve as the pre-culture for Session 2. *Note: Kanamycin is a select carcinogen.***

A) Winter 2024 syllabus excerpt (prior to revisions):

The original lab design created multiple barriers to student success:

- Accessibility issue: Late-night (11 PM) and early-morning (7 AM) requirements created inequitable access for students with employment, family responsibilities, or transportation limitations
- Time management burden: Experiments routinely exceeded the scheduled 4-hour lab period, preventing students from completing their work and disadvantaging those with subsequent classes or commitments
- Pedagogical concern: Time pressure reduced opportunities for in-the-moment instruction, procedural clarification, and collaborative data analysis, diminishing the learning value of hands-on work

B)

Modifications to protocol for session 1:

- No one needs to come in the evening before or 3-4 hours before
- O2 and pH probes are already calibrated
- LB will already have Kanamycin and will be pre-warmed to 37C (reduce lag phase)
- Starter culture will be grown at 25C (so more cells are still growing)
- To start fermentation culture (Step 1 in To do list)
  - Measure OD600
  - Spin down culture and remove media (remove waste products and/or acidified media to reduce lag phase)
  - Add pre-warmed LB-Kanamycin media, get to 1 L of culture at around 0.2-0.4 OD

B) Modifications to protocol during Winter 2024 class:

I redesigned the experimental workflow by:

- Analyzing the timescale and biological constraints of each experimental step
- Consolidating or eliminating preparatory steps that did not compromise experimental integrity
- Restructuring the protocol to fit within the scheduled 4-hour lab period

This led to experiments completed within scheduled time, reduced student stress and better learning (documented in student evaluations).

C)

Important: in the evening before Session 1, leave 1 L of LB-Kanamycin in the 37 °C incubator to pre-warm the media. Inoculate 2 sterile 250 mL flasks filled with 100 mL LB-kanamycin from a starter culture of "pET28a_*lacZ*". Incubate the cultures until the next morning in the shaker at room temperature, 300 rpm. *Note: Kanamycin is a toxin.*

C) Spring 2024 Syllabus Excerpt:

I was invited to teach the course again the following quarter, and the changes were successfully institutionalized.

D)

Step 2: Import Sequence for pET-28a Plasmid

- This should automatically load the plasmid map for pET-28a onto Benchling, as shown to the right.



D) Additional Software Tutorial Excerpt:

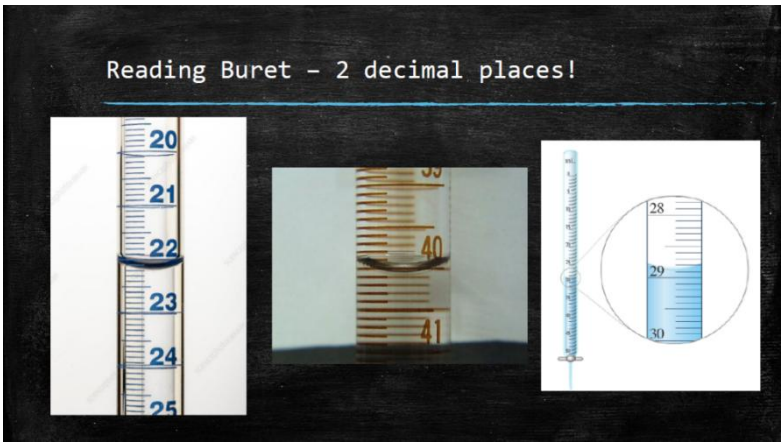
I identified an opportunity to use the reclaimed time during fermentation periods to introduce students to an industry-standard molecular biology software. This new instructional module prepares students for computational skills they will need in biotechnology careers. Initially implemented during my Winter 2024 appointment, this tutorial has now been incorporated as a permanent component of the course.

Teaching Artifact 3: Improving Learning and Accessibility in the Laboratory

CHEM 14BL Selected Lab Section Slides

Context: As students' first college-level chemistry lab experience, many arrive with limited or no prior lab experience. I created visual aids and procedural guides to reduce anxiety, build confidence, and support students in developing foundational lab sk ills.

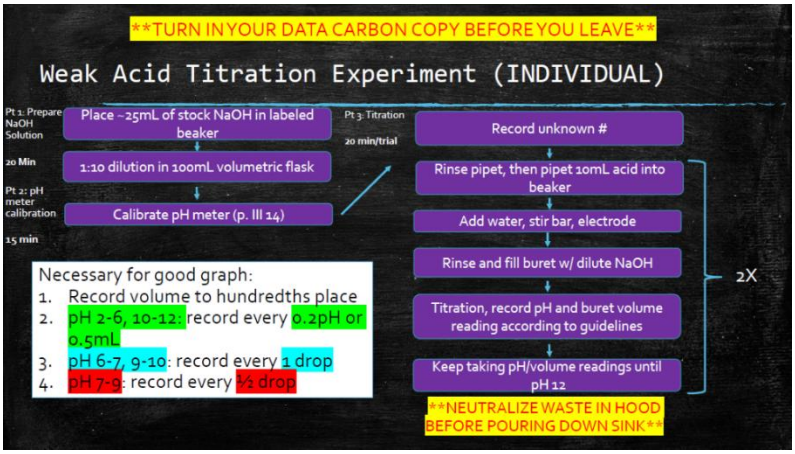
E)



E) Buret reading visual aid

I created a visual aid for reading a buret that allowed students to observe proper meniscus reading. This proactive approach reduced experimental error by building mental models in a low-stakes context, increasing student confidence during hands-on work. By providing explicit instruction on tacit expert knowledge, I helped students develop accurate measurement techniques essential for quantitative analysis throughout the course and in future chemistry and science classes.

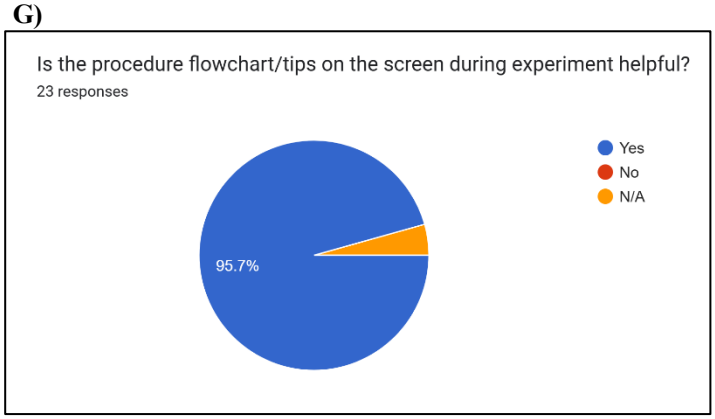
F)

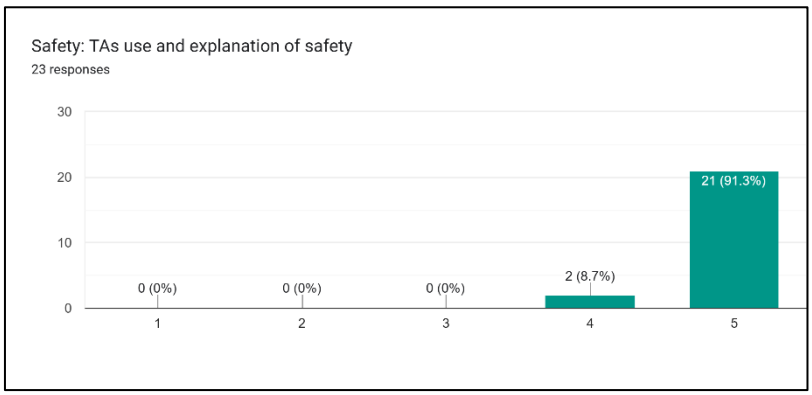


F) Custom simplified lab flowchart

I created a custom flowchart of the lab procedure that remained visible throughout the lab session, allowing students to quickly reference the overall workflow without losing their place in detailed protocol steps. The flowchart highlighted key measurements and integrated critical safety considerations for proper handling and disposal of hazardous chemicals. By providing persistent visual scaffolding, I helped students maintain awareness of both the big picture and safety requirements while reducing the need to constantly flip through the lab manual during critical steps.

G)





G) Mid-quarter evaluation survey results

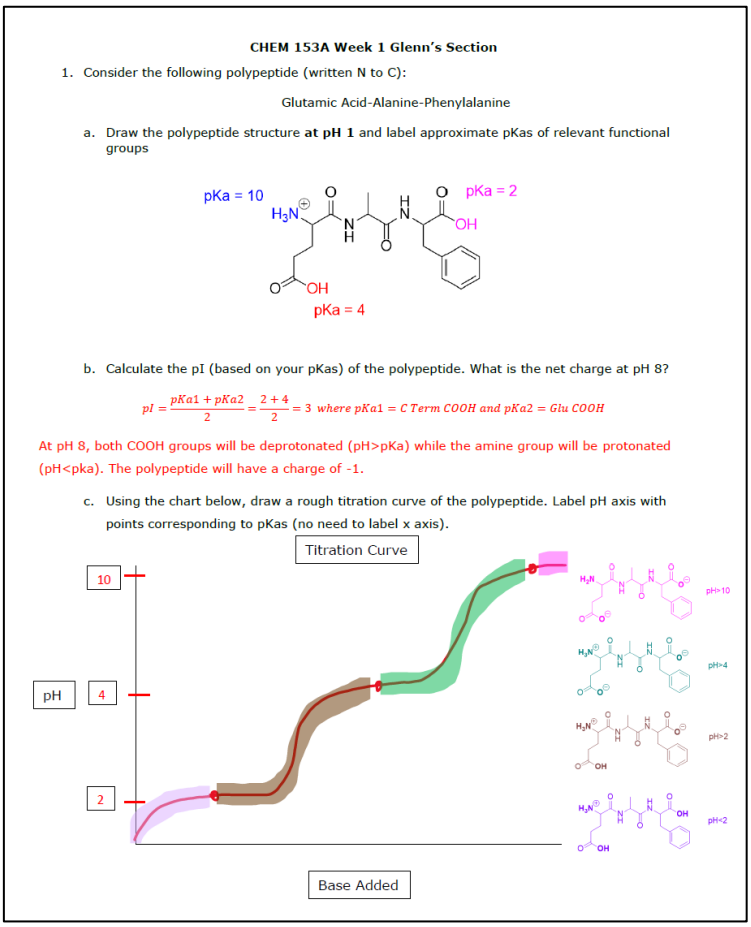
I designed and distributed mid-quarter evaluations on my own initiative to gather feedback that could improve instruction during the remaining weeks of the course. Student responses confirmed the effectiveness of my instructional materials. They emphasized that the flowchart and practical tips were especially helpful and noted feeling more confident due to clear, consistent safety gui delines. While reflective teaching is difficult to implement in real-time especially in the quarter system, my proactive use of recruiting student feedback allowed me to validate which teaching strategies were most impactful and adjust my approach accordingly.

Teaching Artifact 4: Assessment Design for Learning, Transfer, and Exam Preparation

CHEM 153A Discussion Worksheets and Practice Materials

Context: I developed weekly discussion worksheets and comprehensive practice exams to support student learning in this challenging upper-division biochemistry course. These materials went beyond assigned TA duties and were designed to help students build problem-solving skills through formative, low-stakes practice. Students particularly valued that practice questions required critical thinking and integrated multiple concepts, as evidenced in their SET feedback noting these were among the most helpful course resources

A)



CHEM 153A Week 1 Glenn's Section

1. Consider the following polypeptide (written N to C):

Glutamic Acid-Alanine-Phenylalanine

a. Draw the polypeptide structure **at pH 1** and label approximate pKas of relevant functional groups

b. Calculate the pI (based on your pKas) of the polypeptide. What is the net charge at pH 8?

$$pI = \frac{pKa1 + pKa2}{2} = \frac{2+4}{2} = 3 \quad where\ pKa1 = C\ Term\ COOH\ and\ pKa2 = Glu\ COOH$$

At pH 8, both COOH groups will be deprotonated (pH>pKa) while the amine group will be protonated (pH<pka). The polypeptide will have a charge of -1.

c. Using the chart below, draw a rough titration curve of the polypeptide. Label pH axis with points corresponding to pKas (no need to label x axis).

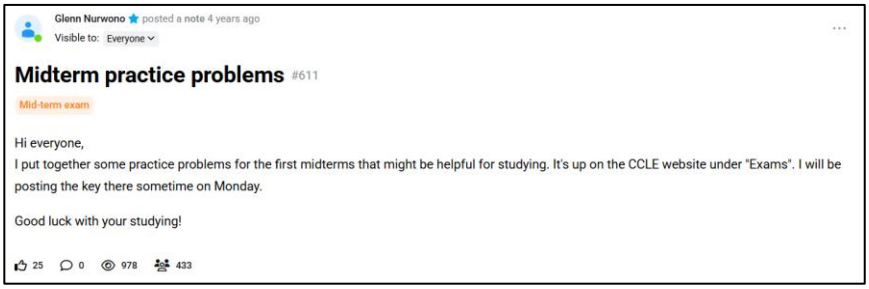
A) Independently developed practice worksheet

I created this worksheet with a complete answer key on my own initiative (beyond assigned TA duties) to help students apply biochemistry concepts through practice problems emphasizing pattern recognition and mechanistic reasoning. Designed for collaborative work during discussion sections, this low-stakes format allowed students to learn from peers, make mistakes safely, and build confidence before high-stakes exams. Student feedback specifically highlighted these practice materials as among the most valuable course resources for exam preparation.

Responses from the mid-quarter evaluation:

- "The resources/practice problems he gives are great for **helping me assess what I know**"
- "I really like that he is **using his own resources to help us better understand concepts** instead of just pulling information off of the professor's slides"

B)



Glenn Nurwono posted a note 4 years ago
Visible to: Everyone

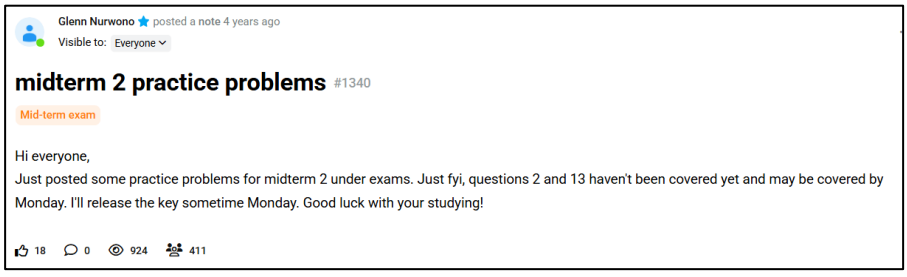
Midterm practice problems #611

Mid-term exam

Hi everyone,
I put together some practice problems for the first midterms that might be helpful for studying. It's up on the CCLE website under "Exams". I will be posting the key there sometime on Monday.

Good luck with your studying!

25 0 978 433



Glenn Nurwono posted a note 4 years ago
Visible to: Everyone

midterm 2 practice problems #1340

Mid-term exam

Hi everyone,
Just posted some practice problems for midterm 2 under exams. Just fyi, questions 2 and 13 haven't been covered yet and may be covered by Monday. I'll release the key sometime Monday. Good luck with your studying!

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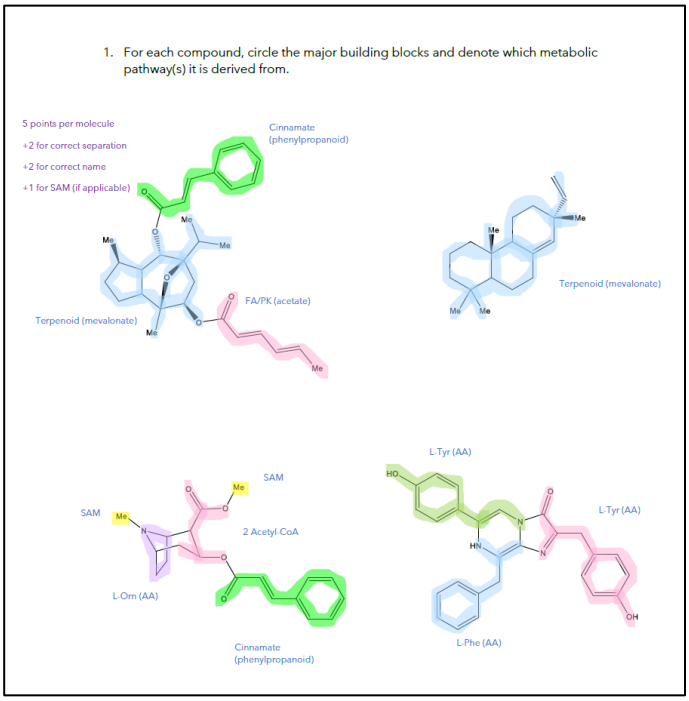
B) Campuswire engagement and course-wide impact. I shared my independently developed practice exams on Campuswire so students across all sections, not just my assigned discussion, could access high-quality study resources. The screenshots show 978 views for the first midterm practice exam and 924 views for the second midterm exam, representing approximately 10 times my section enrollments and reaching nearly all students in the course multiple times. I also actively engaged with student questions on the platform, providing detailed explanations, complete answer keys, and clarifying common misconceptions asynchronously. This proactive sharing extended my teaching impact across the entire course, supported students who could not attend my discussion section, and demonstrated my commitment to equitable access to high-quality learning materials regardless of TA section assignment.

Teaching Artifact 4: Assessment Design for Learning, Transfer, and Exam Preparation

CHEM 138/238 Final Exam Questions

Context: For this inaugural course offering with no prior materials, I played a central role in developing course assessments . I designed final exam questions that emphasized critical thinking and engagement with primary literature, extending learning be yond lecture content into real-world research applications. These questions required students to apply biosynthetic principles to novel scenarios and analyze primary research papers, rather than simply recalling memorized information. This assessment design aligns with evidence-based practices that promote deeper learning and prepare students for research careers .

C)

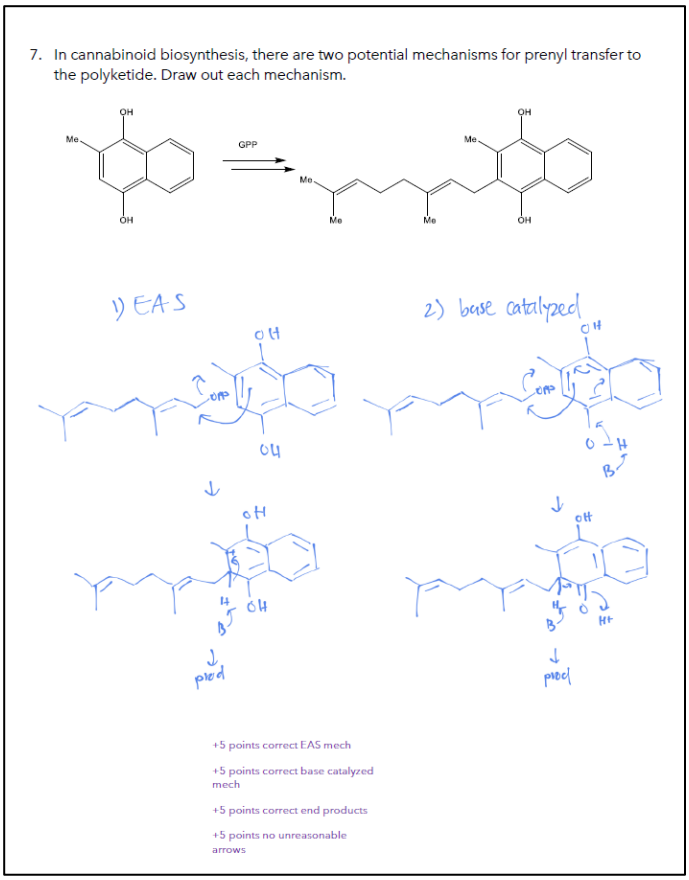


C) Cumulative exam question grounded in primary literature

I designed an exam question based on molecules from the primary literature that required students to synthesize concepts from all course modules by analyzing the biosynthetic origins of each structural component.

Pedagogical Approach: This question tests deeper understanding rather than memorization by requiring students to apply biosynthetic logic to novel molecular structures from actual research papers in contexts they had not encountered in lecture. The multi-part structure with positive scoring and partial credit allows students to demonstrate knowledge across different biosynthetic pathways even if they struggle with one component, reducing all-or-nothing pressure while maintaining intellectual rigor.

D)



D) Scaffolded exam question connecting prior coursework to new research contexts

I wrote an exam question drawing from the instructor's research on cannabinoids that probed students' mechanistic understanding of how these molecules are formed, a core learning objective of the course.

Pedagogical Approach: This question requires transfer of learning, as students must apply organic chemistry reaction mechanisms from prerequisite courses to a more complex biosynthetic context they had not encountered in lecture. Students cannot simply recall worked examples, since they must adapt foundational knowledge to analyze an unfamiliar system. By using cannabinoids from Professor Tang's active research program, the question connects abstract course concepts to real-world scientific inquiry, demonstrating the relevance of biosynthetic analysis to current research problems and increasing student engagement with contemporary science.

Teaching Artifact 5: Student Feedback

Context: Compiled from 7 courses taught across 4 years (Fall 2021 - Winter 2025). This artifact presents student feedback organized thematically to demonstrate consistent patterns in my teaching effectiveness across diverse course levels, formats, and student populations. Quotes are drawn from both introductory labs (CHEM 14BL, first-year students), upper-division lecture discussions (CHEM 153A, juniors/seniors), specialized elective labs (CH ENGR 104D, CHEM 153L), and graduate-level courses (CHEM 138/238, combined undergraduate/graduate). The consistency of themes across contexts demonstrates sustained teaching effectiveness and adaptability.

Approachability & Inclusive Learning Environment

"Genuinely one of the best TA's I have had at UCLA. They were always **super approachable and happy to help**. They really cared about students and their understanding of the material… Knowing my TA was very helpful and **encouraging made me actually look forward to going to class**. Many times I actually found lab to be enjoyable, which is very unusual for me since I have always struggled with chemistry." – CHEM 14BL, Fall 2021

"Literally, Glenn was amazing!! One of my favorite TA's. Chemistry is not everyone's favorite subject and as a requirement, there are a lot of students in the class that this is not their ideal setting, and he did a good job in **making the lab seem less impossible to complete** and made the atmosphere one where **you did not feel stupid asking for help**." – CHEM 14BL, Fall 2021

"Glenn was an amazing TA and I have nothing negative to say about him. He was so **approachable and welcoming** and I am super thankful I had him as my TA. There were times when I felt like I was behind or struggling in lab, and he helped me feel better **and was always there to help**." – CHEM 14BL, Fall 2021

"Glenn really cared about his students learning and made sure that we all felt comfortable coming to him in and outside of class if we ever had any questions about the material. I **always felt welcome coming to him with questions**." – CHEM 14BL, Fall 2021

"I loved having Glenn as a TA this quarter! He is very personable, which makes it **easy to ask questions and have conversations about the material.** This means I always walked away with a better understanding of the material after talking to Glenn. Glenn was very helpful during the lab sections, and very responsive in and out of class. **I would absolutely take another class with Glenn because I know it would be enjoyable and I would learn a lot**!." – CH ENGR 104D, Spring 2024

Impact on Student Learning & Confidence

"Glenn is a great TA, and he really **is eager to help students understand the material**. He's very good at being available to help us with any points of confusion about the labs we are turning in and he **helps us put our best work forward**." – CHEM 14BL, Fall 2021

"Glenn was such a good TA. He was so knowledgeable and had very clear presentations for discussions. I **always felt more confident on class material after attending his sections**. I can tell he cared a lot about how we did in the class because he **always went above and beyond by providing us practice questions and even practice tests** to prepare for the exams." – CHEM 153A, Winter 2022

"Glenn is one of the most amazing TAs I've ever had. He is very well-spoken and evidently knowledgeable about the course content. He was a**ble to make the dense content from lecture digestible for us students to understand**. Further, he is **always prepared**. His slides and practice questions were very organized and super helpful. He was also very approachable. Wonderful TA." – CHEM 153A, Winter 2022

"This was one of the best TA's I have had in a life science course. During every lab period, he was **welcoming, attentive, respectful**. He provided specific presentations during lab each period, **responded to questions through detailed email responses**, and thoroughly described the purpose of experiments and experimental tools to help clarify potentially confusing sections in the lab manual." – CHEM 14BL, Fall 2021

Instructional Innovation & Pedagogical Effectiveness

"Glenn is very knowledgeable about all the course material and is very approachable. In lab he does a **good job of first ensuring that the students know what to expect** (usually with a quick slide show) and also circles around and answers questions/ helps students. Also, he is super good about getting back to and responding in depth to email questions which I really really appreciate!!" – CHEM 14BL, Fall 2021

"Glenn was **really good at explaining the concepts** at the beginning of the section by providing an overview of the mechanisms and providing helpful tips on how to see patterns. I also appreciate that he **made the discussion material available to us online** for those who could not make it to a section or preferred digital resources." – CHEM 153A, Winter 2022

"Glenn **was such a big help in discussion and on campuswire**. He had the **best study questions** and his powerpoints gave a lot of additional help." – CHEM 153A, Winter 2022

Teaching Artifact 5: Student Feedback

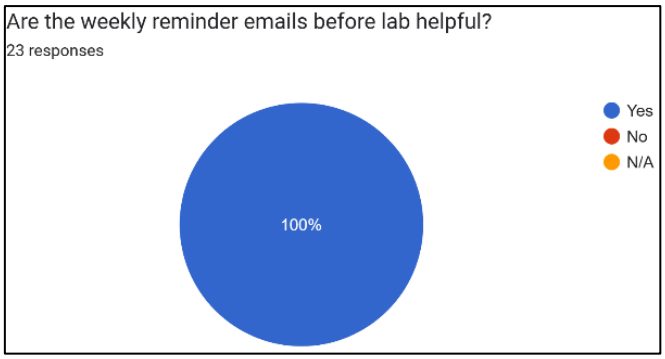
"Glenn was **definitely knowledgeable in the course material** as he assisted Dr. Tang during lecture for any little mishaps the professor would make (if made). The TA also was on top of **providing practice problems to the class** and clarifying in any problem students would have in these worksheets." – CHEM 138/238, Spring 2022

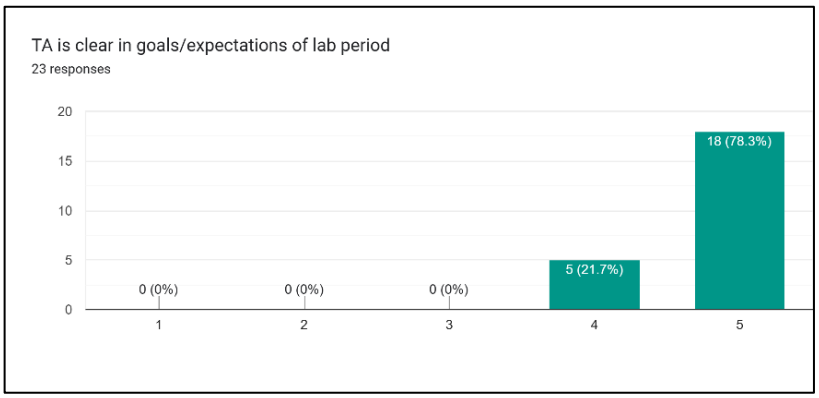
"Glenn is a very knowledgeable and supportive TA. He is **always willing to answer questions** and takes the time to explain concepts clearly**.** His slides are detailed, well–organized, and **make both the paper readings and review lecture very easy to understand**. I don't see any weaknesses in his teaching." – CH ENGR 104D, Fall 2024

"Glenn was a great TA. He always explained the concepts behind our lab very clearly, and provided **very helpful comments on our work in a very short period of time**." – CHEM 153L, Winter 2025

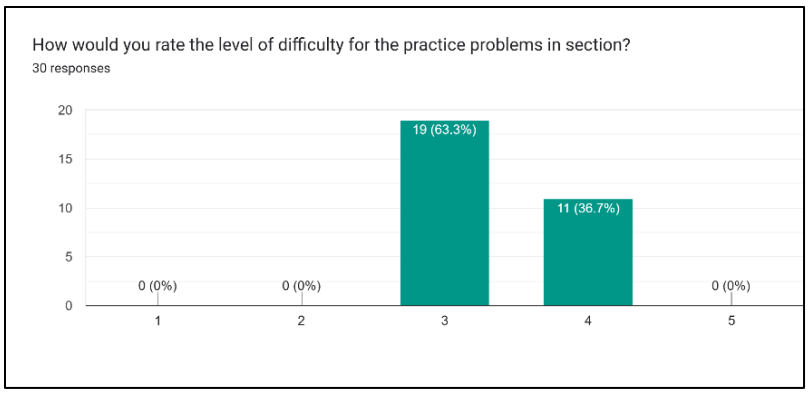
"Glenn was a very good TA who **went the extra mile every week** to create slides for discussion. I liked the format of discussion, with the material being emphasized more than practice problems because biochemistry is just a subject with too much to know. I als o appreciate **Glenn's practice exams, which often required more critical thinking** than the actual exam questions." – CHEM 153A, Winter 2022

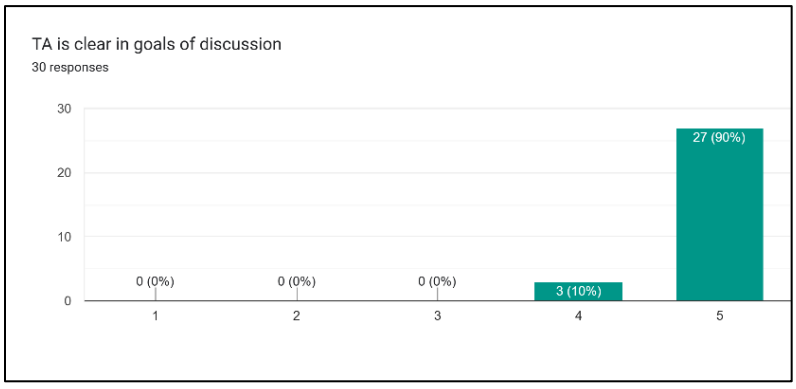
<u>Gauging Student Learning and Preferences Through Mid-Quarter Evaluations:</u>





CHEM 14BL, Fall 2021 (n=23): Results showed that 95% of students rated clarity of lab goals and expectations as "very clear" or "extremely clear," and 100% found weekly reminder emails "helpful" or "very helpful." These data confirm that my visual scaffolding materials (flowcharts, safety guidelines) and proactive communication strategies effectively reduced student anxiety and supported their learning, validating the pedagogical approaches demonstrated in the previous artifact sections.





CHEM 153A, Winter 2022 (n=30):
Results showed that 91% of students rated clarity of discussion goals as "very clear" or "extremely clear," and 87% rated practice problem difficulty as "appropriate" or "challenging in a good way." These findings demonstrate effective communication of learning objectives and successful scaffolding practice materials that prepared students for exams without overwhelming them. This evidence-based approach to formative assessment allowed me to validate and refine my teaching strategies mid-course.

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Glenn has always stood out to me as a natural teacher. Even before he formally was a TA for this course with me, and outside of any assigned TA duties, he was exceptionally helpful during my first time teaching it. Coming from a mammalian cell culture background, I relied heavily on Glenn's expertise in bacterial cell culture protocols and best practices, and he was always eager to teach and support both students and fellow TAs. In the classroom, Glenn's genuine passion for biochemistry and molecular biology is immediately evident. He regularly engages undergraduates at the whiteboards, enthusiastically walking through concepts, encouraging questions, and challenging students to think critically about what they are doing and why. His excitement for teaching is infectious, and students respond strongly to it. Glenn also took initiative to improve a course that had remained largely unchanged for many quarters, modernizing workflows and carefully testing new modules to ensure labs ran smoothly and within scheduled time. His preparation and adaptability consistently created a supportive, well-structured learning environment.

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Glenn is extremely intentional about how students learn. He invests significant time into crafting lesson plans, anticipating student challenges, and adjusting explanations in real time based on student understanding. Rather than allowing lab downtime to go unused, he transformed these moments into learning opportunities. During the fermentation module, for example, he introduced students to molecular biology software to analyze the genetic changes they were implementing and helped them connect abstract concepts to real data while gaining transferable skills relevant beyond the course.

He also actively encouraged discussion around data analysis and lab reports, helping demystify the grading process and showing students how experimental results are interpreted and communicated. Glenn frequently uses whiteboard discussions to break down complex ideas into manageable pieces, reinforcing that learning continues beyond data collection. His approach builds confidence, deepens conceptual understanding, and helps students see themselves as capable scientists.

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name: Justin Purnomo

Briefly describe how you know the nominee. (50 words max)

I engaged with Glenn during my Winter 2022 CHEM 153A course, where he served as a teaching assistant. Although I was not assigned to his section, he kindly provided me with supplementary materials designed for his students, demonstrating his commitment to supporting students' success beyond his assigned responsibilities.

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Throughout the entire course, Glenn consistently prepared high-quality slides and worksheets that I repeatedly found useful. Providing worksheets went above and beyond the expectations for a teaching assistant, and his slides were thorough and effective in consolidating the professor's lectures into a clear, digestible format. His initiative and care for his students inspired me to adopt a similar approach; later, as a graduate student and teaching assistant, I created custom worksheets that were greatly appreciated by my students.

Glenn also looked for ways to improve. I remember him asking me for feedback on the difficulty of his worksheets and midterm review questions. Based on my responses, he

would tailor his discussions to address what he anticipated to be weaker topics. I believe that this proactive approach to teaching allowed him to intellectually challenge students while ensuring they were adequately supported.

Glenn's proactiveness extended beyond the classroom, as he frequently reached out to discuss graduate school with me. As someone without immediate connections in higher education, I found his guidance incredibly helpful. I am positive that many of my classmates benefitted from such interactions, which undoubtedly helped foster trust and relatability between Glenn and his students.

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Glenn's weekly provision of in-depth supplementary materials was incredibly beneficial for my learning and conceptual understanding. These materials were thoughtfully designed to reinforce key concepts based on what he knew about the provided study guides and his past undergraduate biochemistry classes. I regularly referred to his weekly discussion slides and completed his worksheets, eagerly anticipating Fridays when the answer keys were posted to check my work. Glenn's additional midterm review problems were also extremely useful, providing me with an opportunity to test my knowledge in a low-stakes setting. I attribute much of my success in the course to these resources, which played a significant role in helping me earn an A.

This sentiment was shared by many of my peers, as Glenn's dedication to teaching and effectiveness of his materials were frequently praised in our class group chat. I even recall the professor referring to an analogy Glenn made between pH and temperature in relation to enzyme stability, further highlighting the clarity of his teaching.

Overall, Glenn was an open-book who was more than happy to share his knowledge and experience. I can personally attest to the effectiveness of his teaching and give my full recommendation for this award.

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

As mentioned earlier, Glenn has been an invaluable resource in my academic and professional growth. Even after his CHEM 153A responsibilities ended, he was more than willing to answer some of my CHEM 153B questions. During my graduate application cycle, Glenn took the time to review my personal statements and advised me to explore specific chemistry programs that aligned with my research interests in computational protein design and folding. These are only a few of the many personal anecdotes that demonstrate Glenn's dedication and support to go above and beyond for his students, past or present.

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

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P.O. Box 951569
607 Charles E. Young Drive, East
LOS ANGELES, CALIFORNIA 90095-1569

Los Angeles, February 12, 2026

To the UCLA Distinguished Teaching Award Selection Committee:

It is with great enthusiasm that I recommend **Glenn Nurwono** for the **2026 UCLA Distinguished Teaching Award for Teaching Assistants**. Having served as the Course Instructor for Chem 53L since 2014, I have worked with numerous graduate students; however, Glenn's dedication to pedagogical excellence and student success is truly exceptional.

During the Winter Quarter 2025, I witnessed firsthand Glenn's commitment to student learning outcomes in my biochemistry laboratory course (Chem 153L). He is not only a master of complex biochemical techniques but also a gifted communicator of the theoretical principles behind them. His proactive nature is remarkable—when faced with a technique he had not yet mastered, such as SDS-polyacrylamide gel pouring, he sought additional hands-on training outside of class to ensure he could provide high-quality instruction to his students.

Glenn's reliability and dedication extend beyond the classroom. During a building evacuation due to a gas leak, Glenn voluntarily stayed behind after the safety all-clear to help me restore the laboratory, ensuring it was ready for the next day's instruction despite being given the option to leave. This level of professional responsibility is a hallmark of his character.

His teaching portfolio is impressively diverse, spanning Chem 14BL, Chem 153A, Chem C138/238, and Chemical Engineering 104D. In every assignment, he has consistently received excellent student evaluations.

Beyond his required duties, Glenn actively contributes to our teaching culture by attending chemical education research journal clubs and invited seminars.

Notably, Glenn has demonstrated significant initiative in course design and mentorship:

- **Curriculum Innovation:** He designed a writing-intensive course on alternative nutritional proteins, engaging students in multifaceted scientific and social analysis.

- **Mentorship:** Through the IGNITE program, he mentors incoming graduate students in the Department of Chemistry and Biochemistry, helping them refine their own instructional skills.

- **Professional Certification:** He is currently working on his **CIRTL Associate Certification**, demonstrating his mastery of evidence-based teaching, active learning, and inclusive pedagogy.

In summary, Glenn represents the very best of UCLA's teaching community. His blend of technical expertise, pedagogical rigor, and genuine care for student growth makes him an ideal recipient for this prestigious honor. Please feel free to contact me if you require further information.

Sincerely,



Dr. Anne Hong-Hermesdorf
Instructor of Biochemistry